# Endogenous Growth Theory Cheatsheet

Macroeconomics I - Lecture 4

## Contents

# 1 Notation and Definitions

## Core Variables

| | | |
|---|---|---|
| $K$ = capital stock | $L$ = labor force | $Y$ = output |
| $A$ = technology/TFP | $s$ = savings rate | |
| $\delta$ = depreciation rate | $n$ = population growth rate | |
| $M$ = number of varieties (intermediate goods) | $r$ = interest rate | |
| $\rho$ = time preference rate | $\sigma$ = RRA coefficient | |

## Key Parameters

| | |
|---|---|
| $\alpha$ = capital share in production | $\lambda$ = R&D productivity |
| $\eta$ = learning-by-doing externality | $\phi$ = decreasing returns in R&D |

## Key Concepts

- **Non-rival good**: Knowledge used by one agent doesn't preclude use by others
- **Partly excludable**: Patents allow temporary exclusion
- **Monopolistic competition**: Many firms, each with market power in their variety
- **Scale effects**: Growth rate depends on size of economy (L)

## 2 AK Model: Basic Setup

### Production Function

$$Y = AK \tag{AK Production}$$

where $A > 0$ is constant. This is the case $\alpha = 1$ in Cobb-Douglas.

### Capital Accumulation with Exogenous Savings

$$\dot{K} = sY - \delta K = sAK - \delta K \tag{AK Accumulation}$$

Derivation of Growth Rate Dividing by $K$:

$$\gamma_K \equiv \frac{\dot{K}}{K} = sA - \delta$$

Since $Y = AK$, output grows at the same rate:

$$\gamma_Y = \gamma_K = sA - \delta$$

$$\boxed{\gamma = \gamma_Y = \gamma_K = sA - \delta} \tag{AK Growth Rate}$$

**Key Properties**

- No convergence: Growth rate is constant, independent of $K$
- Behavioral parameters affect long-run growth: $s$ directly enters $\gamma$
- Sustained growth requires $sA > \delta$

# 3 AK Model with Learning-by-Doing

### Arrow's Learning-by-Doing

Each firm faces decreasing returns, but aggregate technology has external effect:

$$Y_j = \overline{A} K_j^{\alpha} L_j^{1-\alpha}, \quad \overline{A} = A_0 \left( \int_i K_i \right)^{\eta} \tag{LbD Production}$$

With market clearing $\int_i K_i = K$:

$$Y = A_0 K^{\eta+\alpha} L^{1-\alpha} = A K^{\eta+\alpha}, \quad A \equiv A_0 L^{1-\alpha} \tag{Aggregate LbD}$$

### Growth Regimes

Classification The capital accumulation equation:

$$\dot{K} = sAK^{\eta+\alpha} - \delta K$$

Growth rate:

$$\gamma_K = sAK^{\eta+\alpha-1} - \delta$$

Three cases:

1. If $1 - \alpha - \eta > 0$: Converges to steady state (Solow behavior)
2. If $1 - \alpha - \eta < 0$: Explosive growth (increasing returns)
3. If $1 - \alpha - \eta = 0$: Knife-edge case $\Rightarrow$ sustained growth

$$\boxed{\gamma = sA - \delta \quad \text{when } \eta = 1 - \alpha} \tag{Solow-AK Growth}$$

# 4 AK-Ramsey Model

## Setup

- Same production: $Y = AK$
- CRRA utility: $U = \int_0^\infty \frac{c^{1-\sigma}-1}{1-\sigma} e^{(n-\rho)t} dt$
- Flow budget constraint: $\dot{k}_t = Ak_t - c_t - (\delta + n)k_t$

*Competitive Factor Markets

$$r_t = A - \delta \tag{MPK}$$

Euler Equation Derivation Consumer maximizes utility subject to flow constraint. First-order condition:

$$c_t^{-\sigma} = \beta \int_0^\infty e^{-\int_0^s r_\omega d\omega} c_{t+s}^{-\sigma} ds$$

Differentiating with respect to $t$:

$$\frac{\dot{c}}{c} = \frac{r_t - \rho}{\sigma}$$

$$\boxed{\gamma_c \equiv \frac{\dot{c}}{c} = \frac{A - \delta - \rho}{\sigma}} \tag{AK-Ramsey EE}$$

Interpretation

- Consumption grows at constant rate if $A > \delta + \rho$
- Growth depends on technology $A$, depreciation $\delta$, patience $\rho$, risk aversion $\sigma$
- No scale effects in AK-Ramsey (unlike Romer)

# 5 Externalities and Inefficiency

## Private vs Social Returns

**Private MPK (firm optimizes taking $A$ as given):**

$$R_t = \alpha A K_i^{\alpha-1} L_i^{1-\alpha} \tag{Private MPK}$$

**Social MPK (planner accounts for externality):**

$$R_t^{SP} = (\alpha + \eta) A_0 K^{\alpha+\eta-1} L^{1-\alpha} \tag{Social MPK}$$

Inefficiency Result Private Euler Equation:

$$\gamma_c = \frac{\alpha A_0 K^{\eta} K_i^{\alpha-1} L^{1-\alpha} - \delta - \rho}{\sigma}$$

Social Planner Euler Equation:

$$\gamma_c^{SP} = \frac{(\alpha + \eta) A_0 K^{\alpha+\eta-1} L^{1-\alpha} - \delta - \rho}{\sigma}$$

Since $\eta > 0$: $\gamma_c^{SP} > \gamma_c$

$$\boxed{\text{Social MPK} \ > \ \text{Private MPK} \ \Rightarrow \ \text{too little saving}} \tag{Inefficiency}$$

**Policy Implication**

Positive externality from capital accumulation $\Rightarrow$ market under-allocates resources to savings. Planner accumulates more and grows faster.

# 6 Romer Model (1990): Market Economy

### Final Good Production

$$Y_t = \frac{L^{1-\alpha}}{\alpha} \int_0^{M_t} x_i^\alpha \, di \quad \alpha \in (0,1) \tag{Final Good}$$

### Variety Effect

With symmetric intermediate goods $x_i = x$ for all $i$:

$$Y_t = \frac{M_t^{1-\alpha}}{\alpha} L^{1-\alpha} X_t^\alpha \tag{Variety Effect}$$

More varieties $M_t$ increases productivity (like TFP).

### Intermediate Good Market: Monopolistic Competition

**Inverse Demand (from final good FOC):**

$$p_i = L^{1-\alpha} x_i^{\alpha-1} \tag{Demand}$$

**Monopoly Pricing (with marginal cost $c_i = \alpha$):**

Markup Pricing Monopolist maximizes:

$$\max_{x_i} \pi_i = p_i x_i - c_i x_i = L^{1-\alpha} x_i^\alpha - \alpha x_i$$

FOC: $\alpha L^{1-\alpha} x_i^{\alpha-1} = \alpha$
Solution: $x_i = L$, $p_i = 1$

$$\boxed{\text{Markup } = \frac{1}{\alpha} > 1} \tag{Markup}$$

# 7 Romer Model: Profits and Growth

## Monopoly Profits

Profit Calculation With $x_i = L$:

$$\pi_i = p_i x_i - c_i x_i = (1-\alpha)L$$

All firms earn identical profit: $\pi = (1-\alpha)L$

$$\boxed{\pi = L(1-\alpha)} \qquad \text{(Monopoly Profits)}$$

## Firm Value

Present Value of Perpetual Patent

$$V = \int_t^\infty \pi e^{-r(\tau - t)} d\tau = \frac{\pi}{r} \qquad \text{(Firm Value)}$$

$$\boxed{V = \frac{\pi}{r} = \frac{L(1-\alpha)}{r}} \qquad \text{(Value)}$$

## R&D and Free Entry

$$\dot{M}_t = \lambda R_t \qquad \text{(R\&D Technology)}$$

Research sector profit:

$$\pi^R = \frac{\pi}{r}\lambda R - R$$

Free entry ($\pi^R = 0$):

$$\boxed{r = \lambda\pi = \lambda L(1-\alpha)} \qquad \text{(Equilibrium Interest Rate)}$$

# 8 Romer Model: BGP Growth Rate

### Household Euler Equation

$$\frac{\dot{c}}{c} = \frac{r - \rho}{\sigma} \tag{EE}$$

### Growth on Balanced Growth Path

Market Equilibrium Growth Substitute $r = \lambda L(1 - \alpha)$ into Euler Equation:

$$\gamma \equiv \frac{\dot{c}}{c} = \frac{\lambda(1 - \alpha)L - \rho}{\sigma}$$

GDP growth equals consumption growth:

$$\frac{\partial \log GDP}{\partial t} = \frac{\partial \log M}{\partial t}$$

$$\boxed{\gamma = \frac{\lambda(1 - \alpha)L - \rho}{\sigma}} \qquad \text{(Romer Growth Rate)}$$

**Scale Effects**

- Growth increases with $L$: larger economies grow faster
- Growth increases with R&D productivity $\lambda$
- Growth decreases with time preference $\rho$ and risk aversion $\sigma$
- Empirically rejected: Jones (1995) shows R&D personnel grew 10x since 1950 without growth acceleration

# 9 Romer Model: Planner Solution

### Static Planner Problem

$$\max_{\{x_i\}} \frac{L^{1-\alpha}}{\alpha} \int_0^{M_t} x_i^{\alpha} di - \alpha \int_0^{M_t} x_i \, di \tag{Static Planner}$$

Static Solution FOC: $L^{1-\alpha} x_i^{\alpha-1} = \alpha$
Solve: $x_i^* = L\alpha^{\frac{1}{\alpha-1}} > L = x_i$ (market)
Key difference: Planner uses machines more intensively (no markup)

$$\boxed{x_i^* = L\alpha^{\frac{1}{\alpha-1}} > x_i = L} \tag{Planner Input}$$

### Output Comparison

$$Y_t^* = \frac{M_t}{\alpha} L\alpha^{\frac{\alpha}{\alpha-1}} > Y_t = \frac{M_t L}{\alpha} \tag{Planner Output}$$

$$\tilde{Y}_t^* = \frac{M_t}{\alpha} L\alpha^{\frac{\alpha}{\alpha-1}} (1-\alpha) \tag{Planner Net Output}$$

## 10 Planner vs Market: Growth Comparison

### Dynamic Planner Problem

$$\max_{c_t} \int_0^\infty \frac{c_t^{1-\sigma} - 1}{1-\sigma} e^{-\rho t} dt$$

subject to $\dot{M}_t = \lambda(\tilde{Y}_t^* - c_t)$

Planner Growth Derivation Hamiltonian yields Euler equation:

$$\frac{\dot{c}_t^*}{c_t^*} = \frac{1}{\sigma}\left(\frac{\lambda L \alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} - \rho\right)$$

$$\boxed{\gamma^* = \frac{1}{\sigma}\left(\frac{\lambda L \alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} - \rho\right)} \qquad \text{(Planner Growth)}$$

### Market vs Planner Comparison

Growth Inequality Since $\alpha^{\frac{1}{\alpha-1}} > 1$:

$$\gamma^* = \frac{1}{\sigma}\left(\frac{\lambda L \alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} - \rho\right) > \frac{\lambda(1-\alpha)L - \rho}{\sigma} = \gamma$$

$$\boxed{\gamma^* > \gamma} \qquad \text{(Planner Superiority)}$$

**Policy Implication**

Planner grows faster because:

- No monopoly markup on intermediate goods
- More intensive use of machines: $x_i^* > x_i$
- Optimal subsidy: $\alpha$ to make marginal cost $= \alpha^2/\alpha = \alpha$ (social MC)

## 11 Scale Effects and Jones (1995) Fix

### Scale Effects in Romer

**Problem**

$\gamma = \frac{\lambda(1-\alpha)L-\rho}{\sigma}$ implies:

- Larger economies ($L$) grow faster
- Empirically rejected: R&D personnel grew 10x since 1950, but growth rate stayed constant

### Jones (1995): Decreasing Returns in R&D

$$\dot{M}_t = \lambda M_t^{\phi} L_t^R \quad \text{with } \phi < 1 \tag{Jones R\&D}$$

No-Scale Growth Derivation Researcher's FOC: $L_E \lambda \alpha M^{\phi-1}/r = 1$
On BGP: $L_E/L = \text{const} \Rightarrow \dot{L}_E/L_E = n$
Differentiating FOC:

$$\frac{\dot{L}_E}{L_E} + (\phi - 1)\frac{\dot{M}}{M} = 0$$

Substitute $\dot{L}_E/L_E = n$:

$$\frac{\dot{M}}{M} = \frac{n}{1-\phi}$$

Since GDP grows at rate of varieties:

$$\boxed{\gamma = \frac{n}{1-\phi}} \tag{Jones Growth}$$

**Key Insight**

- No scale effects: $\gamma$ does not depend on $L$
- Growth depends on population growth $n$
- Absent population growth: no endogenous growth
- Spillovers limited ($\phi < 1$) $\Rightarrow$ need population growth for sustained growth

## 12 Model Comparison

### Summary Table

| Model | Growth Engine | Scale Effects | Convergence | Key Feature |
|---|---|---|---|---|
| Solow | Exogenous $g$ | No | Yes | Diminishing returns |
| AK | Capital accum. | No | No | Linear technology |
| AK-Ramsey | Capital accum. | No | No | Endogenous $c$ |
| Romer (1990) | New varieties | Yes | No | Monopolistic competition |
| Jones (1995) | New varieties | No | No | Decreasing returns in R&D |

### Growth Rate Formulas

| Model | Growth Rate $\gamma$ |
|---|---|
| Solow | $g$ (exogenous) |
| AK | $sA - \delta$ |
| AK-Ramsey | $(A - \delta - \rho)/\sigma$ |
| Romer (Market) | $[\lambda(1-\alpha)L - \rho]/\sigma$ |
| Romer (Planner) | $[\lambda L \alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)/\alpha - \rho]/\sigma$ |
| Jones (1995) | $n/(1-\phi)$ |

## 13 Problem-Solving Templates

### (1) AK Model Steady State

1. Write capital accumulation: $\dot{K} = sAK - \delta K$
2. Compute growth rate: $\gamma = \dot{K}/K = sA - \delta$
3. Check if $\gamma > 0$ for sustained growth

### (2) Romer Model: Find Equilibrium

1. Solve final good producer's problem $\rightarrow$ inverse demand
2. Solve monopolist's problem $\rightarrow$ $x_i = L$, $p_i = 1$, $\pi = L(1-\alpha)$
3. Compute firm value: $V = \pi/r$
4. Solve R&D free entry: $r = \lambda\pi$
5. Compute growth from Euler equation

### (3) Planner vs Market Comparison

1. Solve market allocation first (monopolistic competition)
2. Solve planner problem (social marginal cost $= \alpha$)
3. Compare $x_i^*$ vs $x_i$, and growth rates
4. Determine policy: subsidy to eliminate markup

### (4) Scale Effects Analysis

1. Check if $\gamma$ depends on $L$
2. If yes: model has scale effects
3. To fix: introduce $\phi < 1$ in R&D: $\dot{M} = \lambda M^{\phi} L^{R}$
4. Solve for $\gamma = n/(1-\phi)$ (no scale effects)

## 14 Common Pitfalls

1. **Confusing AK and Romer**
   - AK: linear technology $Y = AK$
   - Romer: endogenous variety creation via R&D
2. **Markup formula errors**
   - Markup $= 1/\alpha$, not $1/(1-\alpha)$
   - Price $p = 1$, MC $= \alpha \Rightarrow p/MC = 1/\alpha$
3. **Scale effects confusion**
   - Romer (1990): $\gamma \propto L$ (scale effects)
   - Jones (1995): $\gamma = n/(1-\phi)$ (no scale effects)
4. **Convergence prediction errors**
   - Solow: conditional convergence
   - AK/Romer: no convergence prediction
5. **Planner vs Market**
   - Planner uses machines more intensively: $x_i^* > x_i$
   - Growth: $\gamma^* > \gamma$
6. **Transversality condition**
   - Must hold for optimal allocation: $\lim_{t\to\infty} k_t e^{(n-r)t} = 0$

## 15 Quick Reference Formulas

### AK Model

$$\gamma = sA - \delta \tag{AK}$$

$$\gamma_c = \frac{A - \delta - \rho}{\sigma} \tag{AK-Ramsey}$$

### Romer Model (Market)

$$\begin{aligned}
Y &= \frac{L^{1-\alpha}}{\alpha} \int_0^M x_i^\alpha di \\
x_i &= L, \quad p_i = 1 \\
\pi &= L(1-\alpha) \\
V &= \frac{L(1-\alpha)}{r} \\
r &= \lambda L(1-\alpha) \\
\gamma &= \frac{\lambda(1-\alpha)L - \rho}{\sigma}
\end{aligned}$$

### Romer Model (Planner)

$$\begin{aligned}
x_i^* &= L\alpha^{\frac{1}{\alpha-1}} \\
\gamma^* &= \frac{1}{\sigma}\left(\frac{\lambda L \alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} - \rho\right)
\end{aligned}$$

### Jones (1995) No-Scale

$$\begin{aligned}
\dot{M} &= \lambda M^\phi L^R \\
\gamma &= \frac{n}{1-\phi}
\end{aligned}$$

# 16 Implementation Notes

### Structure

- Sections 1-2: AK Model basics
- Section 3: AK-Ramsey (endogenous consumption)
- Section 4: Externalities and inefficiency
- Sections 5-7: Romer (1990) Market model
- Sections 8-9: Planner comparison
- Section 10: Scale effects and Jones fix
- Sections 11-13: Comparison, templates, pitfalls

### Key Equations to Memorize

- AK growth: $\gamma = sA - \delta$
- AK-Ramsey EE: $\gamma_c = (A - \delta - \rho)/\sigma$
- Romer markup: $1/\alpha$
- Monopoly profit: $\pi = L(1 - \alpha)$
- Interest rate: $r = \lambda L(1 - \alpha)$
- Market growth: $\gamma = (\lambda(1 - \alpha)L - \rho)/\sigma$
- Jones growth: $\gamma = n/(1 - \phi)$

### Packages Used

Same as Solow Cheatsheet: amsmath, amssymb, mathtools, graphicx, xcolor, siunitx, cleveref, hyperref, tcolorbox, booktabs

### Compilation

Compile with: `pdflatex` or `lualatex`